LAB REPORT: SET OPERATIONS, SORTING & RELATIONAL DIVISION

CSBB 204: Database Management Systems

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Aim

To understand and execute complex relational SQL operations: Set Theory operations (`UNION`, `INTERSECT`), multi-level result sorting (`ORDER BY`), derived calculations, and Relational Division using nested correlated subqueries.

Theory

Operation / Concept	Functional Description
`UNION`	Combines the results of two queries into a single distinct result set (eliminates duplicates).
`INTERSECT`	Returns only the distinct rows that appear in the results of both input queries.
`ORDER BY`	Sorts the result set by one or more attributes in ascending (`ASC`, default) or descending (`DESC`) order.
`Relational Division`	Represents queries involving universal quantification ("for all"); implemented in SQL using double `NOT EXISTS` or `GROUP BY ... HAVING COUNT(*)` matching.

1. Projects involving an employee named 'Smith' (as worker or department manager)

```
MariaDB [company]> (SELECT w.pno AS pnumber
    ->  FROM works_on w
    ->  INNER JOIN employee e ON w.essn = e.ssn
    ->  WHERE e.lname = 'Smith')
    -> UNION
    -> (SELECT p.pnumber
    ->  FROM project p
    ->  INNER JOIN department d ON p.dnum = d.dnumber
    ->  INNER JOIN employee e ON d.mgr_ssn = e.ssn
    ->  WHERE e.lname = 'Smith');
+---------+
| pnumber |
+---------+
|       1 |
|       2 |
+---------+
2 rows in set (0.001 sec)
```

2. Names of employees who work on some project OR have at least one dependent

```
MariaDB [company]> SELECT e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN works_on w ON e.ssn = w.essn
    -> UNION
    -> SELECT e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN dependent d ON e.ssn = d.essn;
+----------+---------+
| fname    | lname   |
+----------+---------+
| John     | Smith   |
| Joyce    | English |
| Ramesh   | Narayan |
| Franklin | Wong    |
| Alicia   | Zelaya  |
| Ahmad    | Jabbar  |
| Jennifer | Wallace |
| James    | Borg    |
+----------+---------+
8 rows in set (0.001 sec)
```

3. Employees who either manage a department OR have at least one dependent

```
MariaDB [company]> SELECT e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN department d ON e.ssn = d.mgr_ssn
    -> UNION
    -> SELECT e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN dependent dp ON e.ssn = dp.essn;
+----------+---------+
| fname    | lname   |
+----------+---------+
| James    | Borg    |
| Robert   | Scott   |
| Jennifer | Wallace |
| John     | Smith   |
| Franklin | Wong    |
+----------+---------+
5 rows in set (0.001 sec)
```

4. Employees who are BOTH department managers AND work on some project

```
MariaDB [company]> SELECT e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN department d ON e.ssn = d.mgr_ssn
    -> INTERSECT
    -> SELECT e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN works_on w ON e.ssn = w.essn;
+----------+---------+
| fname    | lname   |
+----------+---------+
| John     | Smith   |
| Jennifer | Wallace |
| James    | Borg    |
+----------+---------+
3 rows in set (0.001 sec)
```

5. Employees who BOTH work on projects AND have at least one dependent

```
MariaDB [company]> SELECT e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN works_on w ON e.ssn = w.essn
    -> INTERSECT
    -> SELECT e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN dependent d ON e.ssn = d.essn;
+----------+---------+
| fname    | lname   |
+----------+---------+
| John     | Smith   |
| Franklin | Wong    |
| Jennifer | Wallace |
+----------+---------+
3 rows in set (0.001 sec)
```

6. Employees and their projects ordered by department, then last name, then first name

```
MariaDB [company]> SELECT d.dname, e.lname, e.fname, p.pname
    -> FROM employee e
    -> INNER JOIN department d ON e.dno = d.dnumber
    -> INNER JOIN works_on w ON e.ssn = w.essn
    -> INNER JOIN project p ON w.pno = p.pnumber
    -> ORDER BY d.dname ASC, e.lname ASC, e.fname ASC;
+----------------+---------+----------+-----------------+
| dname          | lname   | fname    | pname           |
```

```
+----------------+---------+----------+-----------------+
| Administration | English | Joyce    | ProductX        |
| Administration | English | Joyce    | ProductY        |
| Administration | Wallace | Jennifer | Reorganization  |
| Administration | Wallace | Jennifer | Allbenefits     |
| Administration | Zelaya  | Alicia   | Computerization |
| Administration | Zelaya  | Alicia   | Allbenefits     |
| Headquarters   | Borg    | James    | Reorganization  |
| Headquarters   | Jabbar  | Ahmad    | Computerization |
| Headquarters   | Jabbar  | Ahmad    | Allbenefits     |
| Research       | Narayan | Ramesh   | ProductZ        |
| Research       | Smith   | John     | ProductX        |
| Research       | Smith   | John     | ProductY        |
| Research       | Wong    | Franklin | Computerization |
| Research       | Wong    | Franklin | Reorganization  |
+----------------+---------+----------+-----------------+
14 rows in set (0.001 sec)
```

7. All employees with salary ordered by department number and descending salary

```
MariaDB [company]> SELECT dno, fname, lname, salary
    -> FROM employee
    -> ORDER BY dno ASC, salary DESC;
+-----+----------+-----------+----------+
| dno | fname    | lname     | salary   |
+-----+----------+-----------+----------+
|   1 | Robert   | Scott     | 70040.00 |
|   1 | James    | Borg      | 65900.00 |
|   1 | Ahmad    | Jabbar    | 24500.00 |
|   1 | Anirudh  | Periasamy |     NULL |
|   4 | Jennifer | Wallace   | 49340.00 |
|   4 | Alicia   | Zelaya    | 24500.00 |
|   4 | Joyce    | English   | 24500.00 |
|   5 | Franklin | Wong      | 45200.00 |
|   5 | Ramesh   | Narayan   | 42440.00 |
|   5 | John     | Smith     | 31400.00 |
+-----+----------+-----------+----------+
10 rows in set (0.000 sec)
```

8. Employees and projects ordered by project number, then employee last name

```
MariaDB [company]> SELECT p.pnumber, p.pname, e.lname, e.fname
    -> FROM project p
    -> INNER JOIN works_on w ON p.pnumber = w.pno
    -> INNER JOIN employee e ON w.essn = e.ssn
    -> ORDER BY p.pnumber ASC, e.lname ASC;
+---------+-----------------+---------+----------+
| pnumber | pname           | lname   | fname    |
+---------+-----------------+---------+----------+
|       1 | ProductX        | English | Joyce    |
|       1 | ProductX        | Smith   | John     |
|       2 | ProductY        | English | Joyce    |
|       2 | ProductY        | Smith   | John     |
|       3 | ProductZ        | Narayan | Ramesh   |
|      10 | Computerization | Jabbar  | Ahmad    |
|      10 | Computerization | Wong    | Franklin |
|      10 | Computerization | Zelaya  | Alicia   |
|      20 | Reorganization  | Borg    | James    |
|      20 | Reorganization  | Wallace | Jennifer |
|      20 | Reorganization  | Wong    | Franklin |
|      30 | Allbenefits     | Jabbar  | Ahmad    |
|      30 | Allbenefits     | Wallace | Jennifer |
|      30 | Allbenefits     | Zelaya  | Alicia   |
+---------+-----------------+---------+----------+
14 rows in set (0.001 sec)
```

9. Dependents ordered by parent employee last name and dependent name

```
MariaDB [company]> SELECT e.lname AS parent_lname, e.fname AS parent_fname,
    ->        d.dependent_name, d.relationship
    -> FROM employee e
    -> INNER JOIN dependent d ON e.ssn = d.essn
    -> ORDER BY e.lname ASC, d.dependent_name ASC;
+--------------+--------------+----------------+--------------+
| parent_lname | parent_fname | dependent_name | relationship |
+--------------+--------------+----------------+--------------+
| English      | Joyce        | John           | SPOUSE       |
| Scott        | Robert       | Ahmad          | SON          |
| Smith        | John         | Alice          | Daughter     |
| Smith        | John         | Elizabeth      | Spouse       |
| Smith        | John         | Michael        | Son          |
| Wallace      | Jennifer     | Abner          | Spouse       |
| Wong         | Franklin     | Alice          | Daughter     |
| Wong         | Franklin     | Joy            | Spouse       |
| Wong         | Franklin     | Theodore       | Son          |
+--------------+--------------+----------------+--------------+
9 rows in set (0.001 sec)
```

10. Projected salaries with a 10% raise for employees working on 'ProductX'

```
MariaDB [company]> SELECT DISTINCT e.fname, e.lname, e.salary AS current_salary,
    ->        ROUND(e.salary * 1.10, 2) AS raised_salary
    -> FROM employee e
    -> INNER JOIN works_on w ON e.ssn = w.essn
    -> INNER JOIN project p ON w.pno = p.pnumber
    -> WHERE p.pname = 'ProductX';
+-------+---------+----------------+---------------+
| fname | lname   | current_salary | raised_salary |
+-------+---------+----------------+---------------+
| John  | Smith   |       31400.00 |      34540.00 |
| Joyce | English |       24500.00 |      26950.00 |
+-------+---------+----------------+---------------+
2 rows in set (0.000 sec)
```

11. Names of employees who work on all projects controlled by Department 5 (Relational Division)

```
MariaDB [company]> SELECT e.fname, e.lname
    -> FROM employee e
    -> WHERE NOT EXISTS (
    ->     SELECT p.pnumber
    ->     FROM project p
    ->     WHERE p.dnum = 5
    ->     AND NOT EXISTS (
    ->         SELECT *
    ->         FROM works_on w
    ->         WHERE w.essn = e.ssn AND w.pno = p.pnumber
    ->     )
    -> );
+-------+-------+
| fname | lname |
+-------+-------+
| John  | Smith |
+-------+-------+
1 row in set (0.001 sec)
```